StegaGuard: A Controlled Robustness Study of Lightweight Image Steganalysis under Algorithm and Payload Shift

Quoc Hung NGUYEN *[0000-0002-6363-0160], Minh Tri HUYNH [0009-0004-4214-6355], Phu NGUYEN [0009-0004-7291-9072], Huu Minh Tri NGUYEN [0009-0002-8275-4326], Duong Gia Bao NGUYEN [0009-0009-3717-2424], Huu Loc TRAN [0009-0004-0634-0310]

University of Economics Ho Chi Minh City (UEH), Ho Chi Minh City, Vietnam
Corresponding author: hungngq@ueh.edu.vn

Abstract. Steganalysis detectors are trained on known embedding algorithms and payloads, but an adversary can change both. StegaGuard is a controlled robustness study of two existing feature-based detectors: Residual-LR (76 residual statistics, logistic regression) and SRM+Ensemble (34,671 Spatial Rich Model features, Fisher linear discriminant ensemble). We split 600 BOSSbase 1.01 covers, resized to 256 × 256, into 360 training, 120 validation and 120 test covers before embedding them with WOW, S-UNIWARD and HILL at 0.1 and 0.4 bpp, so every stego variant stays in the partition of its source cover. Three training protocols of 360 cover-stego pairs were run with three seeds, and paired differences were assessed by a bootstrap over test covers. Trained on WOW at 0.4 bpp, SRM+Ensemble reached an AUC of 0.714 on matched WOW and 0.645 on unseen HILL (ΔAUC = −0.070, 95% CI [−0.093, −0.046]). Adding 0.1 bpp variants to a WOW/S-UNIWARD training pool gave SRM+Ensemble no reliable change on unseen HILL (ΔAUC = −0.022, 95% CI [−0.050, +0.005]). AUC at 0.1 bpp was lower than at 0.4 bpp in all 18 comparisons (ΔAUC −0.022 to −0.183). At thresholds set for a validation false-positive rate (FPR) of at most 1%, mean recall was 0.062 for Residual-LR and 0.080 for SRM+Ensemble at mean test FPRs of 0.045 and 0.034, so neither detector operated reliably at a 1% false-alarm budget on this resized BOSSbase subset.

Keywords: Image Steganalysis, Robustness Evaluation, Algorithm Mismatch, Payload Mismatch, Spatial Rich Model, Low False-Positive Rate.

1 Introduction

Image steganography hides a message by making small changes to a cover image. Current spatial-domain schemes such as WOW, S-UNIWARD and HILL assign a distortion cost to every pixel and use syndrome-trellis coding, or a simulation of it, to place most changes in textured regions that are hard to model [1-4]. Steganalysis detectors either train a classifier such as the Fisher linear discriminant ensemble on noise-residual statistics like Spatial Rich Model (SRM) features, or train a convolutional neural network (CNN) such as SRNet or Yedroudj-Net [5-7].

In a security setting the detector does not choose its test data: a defender trains on a few algorithms at known payloads, while an adversary may use an unseen algorithm or

a shorter message. The operating point matters as well: a screening system inspects many innocent images, so what counts is the share of stego images caught when the false-positive rate (FPR) is held near 1%, not only the area under the ROC curve (AUC).

Both issues are known, but they are usually studied one at a time. Work on cover-source mismatch changes where the images come from [8, 9], work on stego-source mismatch changes the embedding method [10, 11], and papers that introduce detectors mostly train and test on the same algorithm and payload [5, 7, 12]. Two design choices can also bias robustness claims: splitting after embedding lets variants of one cover fall on both sides and inflate test scores, and a detector that is weak in the matched condition says little about the shift itself.

The gap has two parts. For the detector, it is unclear how much a standard SRM+Ensemble loses on a small, resized BOSSbase subset when the algorithm or payload changes, and whether mixing algorithms and payloads in training reduces the loss. For the evaluation, few shift studies combine a source-cover split fixed before embedding, a matched-condition reference, paired intervals on identical test covers and an explicit low-FPR operating point. StegaGuard (Fig. 1) addresses the second part to answer the first; it evaluates two existing detectors and proposes no new detection algorithm. Three questions guide the study:

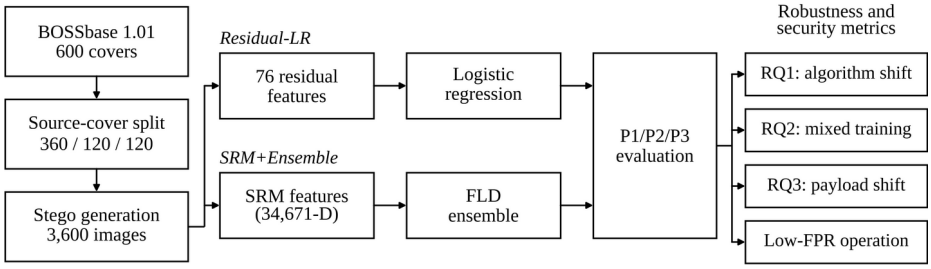


Fig. 1. StegaGuard evaluation pipeline. Both detectors are trained on the same splits, protocols and seeds, and every model is scored on the same 120 test covers.

- **RQ1 (unseen algorithm).** Under protocol P1, which trains on WOW at 0.4 bits per pixel (bpp), how does detection of unseen HILL at 0.4 bpp compare with detection of matched WOW at 0.4 bpp on the same test covers?
- **RQ2 (mixed training).** With training size fixed at 360 cover-stego pairs, does adding 0.1 bpp variants to a WOW/S-UNIWARD training pool (P3 versus P2) improve detection of unseen HILL at 0.4 bpp?
- **RQ3 (payload shift).** For each detector, protocol and algorithm, how much does detection change when the payload falls from 0.4 to 0.1 bpp?

Contributions. (i) A controlled evaluation in which both detectors share covers, training size, seeds and test images, so that differences between protocols P1-P3 trace back to the training pool. (ii) Source-cover leakage prevention: the split is fixed before embedding and audited against a hashed manifest. (iii) A robustness assessment of

algorithm and payload shift and of mixed training with paired ΔAUC and bootstrap intervals over test covers. (iv) A security operating-point analysis that separates the validation-selected threshold from the FPR observed on test covers.

2 Related Work

2.1 Image Steganalysis Techniques

Content-adaptive embedding places changes where the cover is noisy. WOW derives its costs from directional wavelet residuals, S-UNIWARD from the relative change of a wavelet decomposition, and HILL from a high-pass residual smoothed by low-pass filters [2-4]. Because the three cost functions favor different structures, a detector trained on one algorithm meets a partly different change pattern when tested on another. Feature-based steganalysis models the noise residual: SPAM uses Markov transition probabilities of pixel differences [13], and SRM gathers co-occurrences of many linear and non-linear residuals into 34,671 features [12], usually classified by the Fisher linear discriminant (FLD) ensemble, which handles this dimension through random subspaces and bagging [6]. CNN detectors such as Yedroudj-Net and SRNet report lower error than SRM with the ensemble in most tested BOSSbase-derived settings, at a higher training cost [5, 7].

2.2 Robustness under Algorithm and Payload Shifts

Three kinds of train-test shift should be kept apart. *Algorithm shift*, or stego-source mismatch, occurs when test stego images come from an embedding method absent from training. Butora and Fridrich trained SRNet on three of WOW, S-UNIWARD, HILL and MiPOD at 0.4 bpp and tested on the fourth; the largest loss occurred when HILL was held out, while training on several algorithms cost little against dedicated detectors [10]. Recent work adapts to algorithm and cover-source mismatch with unlabeled target data [11]. *Payload shift* occurs when the message length differs between training and test. Detection error grows as the payload falls; SRNet built its low-payload detectors by curriculum training across payloads [5], and knowledge distillation has been proposed for test payloads unseen in training [14]. *Cover-source shift* arises when test covers come from other cameras or processing pipelines and can raise the error even when algorithm and payload match [8, 9]. This study varies the algorithm and the payload and holds the cover source fixed.

Robustness also depends on the operating point. The ALASKA challenge ranked detectors by the missed-detection rate at 5% false alarms (MD5) [15], and the FLD ensemble has been extended to meet a prescribed false-alarm probability [16]. Butora and Bas reported that two detectors with almost equal error differed by about ten points in detection rate at an FPR of 10^{-3}, while noting that such estimates from 2,000 test images are noisy [17].

2.3 Research Gap and Comparative Analysis

Table 1 compares representative studies; "Not reported" means the paper does not report the item.

Table 1. Comparison with representative steganalysis studies. P_E: minimum average error under equal priors; P_{MD}: missed-detection probability; CSM: cover-source mismatch.

Study	Data	Detector	Protocol and metric	Shift	Low-FPR metric
SRM [12]	BOSSbase 0.92, 512 × 512	SRM + FLD ensemble	Matched algorithm and payload; P_E	None	Not reported
Yedroudj-Net [7]	BOSSbase 1.01, resized to 256 × 256	CNN	WOW, S-UNIWARD 0.2/0.4 bpp; matched; P_E	None	Not reported
SRNet [5]	BOSSbase 1.01 + BOWS2, 256 × 256	Residual CNN	0.1-0.5 bpp, curriculum over payloads; P_E	None	Not reported
CSM [9]	Several cover sources, spatial and JPEG	Feature-based, various dimensions	Train and test on different sources	Cover source	Not reported
Butora & Fridrich [10]	BOSSbase 1.01 + BOWS2, 256 × 256	SRNet variants	Leave one of 4 algorithms out, 0.4 bpp; P_{MD}	Algorithm	Not reported
ALASKA [15]	JPEG images with diverse processing	Challenge entries	Challenge ranking	Cover source, algorithm	MD5 (5% FPR)
MODE [17]	BOSSbase, 512 × 512	Pretrained SRNet	HILL 0.3 bpp; P_E, detection at fixed FPR	None	FPR 10^{-3}-10^{-1}
This study	BOSSbase 1.01, 600 covers, 256 × 256	Residual-LR; SRM+Ensemble	Split before embedding; paired bootstrap	Algorithm, payload	Recall, test FPR at 1% val. FPR

Detector papers evaluate the matched condition, shift studies mostly concern the cover source or, for algorithm shift, use large CNN training sets [10], and low-FPR evaluation is rarely combined with algorithm or payload shift [15, 17]. To our knowledge, no study measures algorithm and payload shift for feature-based detectors under a split fixed before embedding, with paired intervals on identical test covers, while reporting the FPR actually reached at a validation-selected threshold. StegaGuard addresses this gap.

3 Methodology

Fig. 1 shows the pipeline. The left column builds a leakage-controlled dataset, and the two middle lanes train both detectors in parallel under protocols P1-P3. The right side evaluates every trained model on one fixed set of test covers.

3.1 Dataset Preparation and Leakage Prevention

BOSSbase 1.01 contains 10,000 grayscale images of 512 × 512 pixels [18]. We drew 600 of them (6%) with the fixed seed 20260918 and bicubically downsampled each to 256 × 256 before embedding, as several CNN studies do [5, 7]. Resizing alters cover noise statistics, so our numbers are not comparable with results at 512 × 512.

Leakage is prevented by the order of operations. The 600 source covers were first assigned to training (360), validation (120) and test (120). Only then was each cover embedded with WOW [2], S-UNIWARD [3] and HILL [4] at 0.1 and 0.4 bpp using the conseal embedding simulator, giving six variants per cover: 600 stego images per algorithm-payload condition, 3,600 in total and 4,200 PNG files with the covers. Training, validation and test hold 2,160, 720 and 720 stego images (360, 120 and 120 per condition). Each stego file inherits the cover_id and split of its source, so a cover and its six variants always share a partition. The manifest records cover_id, split, algorithm, payload, embedding seed and SHA-256 hash; a manifest audit confirmed that every stego variant stays in the partition of its source cover (manifest SHA-256 13c01ca9…2799e8c).

3.2 Feature Extraction and Detectors

Both detectors output a score, higher for stego, that is used for every metric; Table 2 summarizes their configuration. Residual-LR, the lightweight baseline, describes each image by 76 statistics of its noise residuals and classifies them with scikit-learn logistic regression [19]. The residuals are the first- and second-order horizontal and vertical pixel differences. Each gives a normalized 17-bin histogram of its values clipped to [−8, 8] plus the mean absolute value and standard deviation of the unclipped residual (4 × 19 = 76 features). A StandardScaler fitted only on each run's training features standardizes all splits. The classifier uses L2 regularization with C = 1.0, the deterministic lbfgs solver, no class weights and max_iter = 2000. Its decision function is the score. SRM+Ensemble, the established reference, extracts the 106 SRM submodels (34,671 features) [12] with the sealwatch library [20] and trains the FLD ensemble of [6] with L = 50 base learners and random feature subspaces of dimension d_{sub} = 500.

Table 2. Detector configuration.

Setting	Residual-LR	SRM+Ensemble
Features	76 residual statistics	SRM: 106 submodels, 34,671 dimensions
Classifier	logistic regression (scikit-learn)	FLD ensemble, L = 50, d_{sub} = 500
Training data per run	360 cover-stego pairs	360 cover-stego pairs
Seed-dependent parts	variant sampling (P2, P3)	variant sampling (P2, P3); random parts of model fitting

3.3 Experimental Protocols

The protocols in Table 3 differ only in the stego variant attached to each of the 360 training covers. For every training cover and seed, one variant is sampled from the protocol's pool, so each run sees 360 cover-stego pairs built from the same 360 covers; P1 involves no sampling. Validation images serve only to set thresholds. HILL is never used for model fitting or threshold selection, and S-UNIWARD is unseen under P1. Seeds 42-44 control the variant draws and the random parts of SRM+Ensemble fitting.

Table 3. Training protocols. All share the same 360 training, 120 validation and 120 test covers; every model is tested on all six conditions (3 algorithms × 2 payloads).

P	Training stego pool	Variants in pool	Role
P1	WOW 0.4 bpp	1	RQ1 (WOW 0.4 matched, HILL 0.4 unseen); RQ3
P2	WOW and S-UNIWARD at 0.4 bpp	2	RQ2 reference; RQ3
P3	WOW and S-UNIWARD at 0.1 and 0.4 bpp	4	RQ2 (P3 − P2 on HILL 0.4); RQ3

RQ1 uses P1 because it trains on WOW 0.4 alone, so WOW 0.4 is matched and HILL 0.4 unseen for the same model. P2 and P3 differ only in the 0.1 bpp variants, which RQ2 isolates. In every comparison both conditions are scored on the same 120 test covers.

3.4 Evaluation Metrics

For each run and test condition a detector produces 240 scores (120 test covers and their 120 stego variants), from which ROC-AUC is computed. Two thresholds are fixed from validation scores alone, never from test labels. The first, τ_{BER}, is the balanced-error point on the validation scores, with BER = (FPR + P_{MD})/2 and P_{MD} the missed-detection rate; F1 = 2·Prec·Rec/(Prec + Rec) and BER are reported on test at this threshold. The second, τ_{low}, defines the security operating point. It is set on the set V of 120 validation covers so that, with s(c) the score of cover c,

$$FPR_V(\tau_{low}) = \frac{\left|\{c \in V : s(c) \geq \tau_{low}\}\right|}{|V|} \leq 0.01 \quad (1)$$

With |V| = 120, this allows at most one validation false positive (0.83%). At τ_{low} we report recall, the share of the 120 test stego images with $s \geq \tau_{low}$, and the observed test FPR, the share of the 120 test covers with $s \geq \tau_{low}$. The first is the quantity of security interest; the second shows whether the 1% target transfers from validation to test, which Eq. (1) does not guarantee.

3.5 Statistical Analysis and Reproducibility

Each comparison is a paired difference ΔAUC = AUC(shifted) − AUC(reference): HILL 0.4 minus WOW 0.4 under P1 (RQ1), P3 minus P2 on HILL 0.4 (RQ2), and 0.1 minus 0.4 bpp for each detector, protocol and algorithm (RQ3, 18 comparisons). Negative values mean a loss under the shift. The point estimate is the mean of the three per-seed differences, which equals the difference of the unrounded seed-mean AUCs (rounded table values may differ by 0.001). Its 95% interval comes from a paired bootstrap in which the resampling unit is the source test cover: each replicate draws 120 test cover IDs with replacement, and a drawn cover brings all of its cover and stego scores under both compared conditions, so the pairing is preserved. Individual stego images are never resampled on their own. We resample the 120 test covers B = 1,000

times with seed 2026, so every condition and run uses the same resampled cover IDs. The 2.5th and 97.5th percentiles give the 95% interval.

Seed standard deviations (at most 0.028, Supplementary Table S1) describe training randomness on fixed data; the bootstrap describes the sampling of test covers given the trained models. Neither reflects a different choice of covers or split, so the intervals are conditional on this partition: they account only for test-cover sampling and do not extend to other image sources. A direction is called supported only when its interval excludes zero; an interval containing zero is inconclusive, not evidence of equivalence. The 18 RQ3 intervals are not adjusted for multiplicity. Each run saves its per-image test scores and validation thresholds, and the manifest records the embedding seed and SHA-256 hash of every image. All tables can be recomputed from these files.

4 Results and Discussion

4.1 Overall Detection Performance

Table 4 gives the ROC-AUC for every detector, protocol and test condition; F1, BER, recall and seed standard deviations are in Supplementary Table S1. All cells share the 120 test covers, so they are paired observations, not independent experiments. Residual-LR stays close to chance, with AUC of 0.500-0.570 and BER of at least 0.438. SRM+Ensemble separates covers from stego at 0.4 bpp (AUC 0.632-0.717, BER 0.347-0.400) but hardly at 0.1 bpp (0.517-0.550). As Residual-LR fails even in the matched condition, SRM+Ensemble carries the main evidence for the research questions.

Table 4. ROC-AUC (mean over seeds 42-44) on the 120 test covers at 0.4 and 0.1 bpp. Bold: the matched condition of P1. HILL is unseen under all protocols; S-UNIWARD is unseen under P1.

Detector	P	WOW		S-UNIWARD		HILL	
		0.4	**0.1**	**0.4**	**0.1**	**0.4**	**0.1**
Residual-LR	P1	**0.546**	0.510	0.565	0.505	0.570	0.514
	P2	0.534	0.500	0.549	0.504	0.551	0.505
	P3	0.523	0.501	0.555	0.502	0.546	0.511
SRM+Ensemble	P1	**0.714**	0.539	0.689	0.532	0.645	0.525
	P2	0.713	0.550	0.717	0.534	0.653	0.534
	P3	0.686	0.546	0.677	0.528	0.632	0.517

4.2 RQ1: Unseen Embedding Algorithm

Under P1, SRM+Ensemble falls from 0.714 on matched WOW 0.4 to 0.645 on unseen HILL 0.4 (Table 5); the interval of the paired ΔAUC excludes zero, and BER rises from 0.347 to 0.386. For Residual-LR the difference is +0.024 [−0.007, +0.056], but with a matched AUC of only 0.546 the matched signal is too weak to serve as a reference for a shift.

This loss need not be a pure effect of algorithm mismatch. HILL was designed to be harder to detect than WOW with rich-model features [4], so part of the gap may reflect its intrinsic difficulty; the P1 ordering WOW 0.714 > S-UNIWARD 0.689 > HILL 0.645 fits both explanations. Separating them needs a matched HILL reference (trained and tested on HILL), which this study did not run.

Answer to RQ1. On this subset, SRM+Ensemble trained on WOW at 0.4 bpp detects unseen HILL at 0.4 bpp less well than matched WOW on the same test covers (ΔAUC = −0.070, 95% CI [−0.093, −0.046]). This is a measured degradation under the tested shift; it does not establish robustness, or a general lack of it, for other algorithms, payloads or cover sources.

Table 5. Paired comparisons for RQ1 and RQ2 on the same 120 test covers. ΔAUC is computed from unrounded scores and may differ by 0.001 from the difference of the rounded AUCs.

Detector	Reference AUC	Shifted AUC	ΔAUC	95% CI	Reading
RQ1: matched WOW 0.4 → unseen HILL 0.4, both under P1					
Residual-LR	0.546	0.570	+0.024	[−0.007, +0.056]	Weak matched signal
SRM+Ensemble	0.714	0.645	−0.070	[−0.093, −0.046]	Loss, CI excludes 0
RQ2: P2 → P3, both tested on unseen HILL 0.4					
Residual-LR	0.551	0.546	−0.005	[−0.020, +0.007]	Inconclusive
SRM+Ensemble	0.653	0.632	−0.022	[−0.050, +0.005]	Inconclusive

4.3 RQ2: Mixed Training

P3 adds 0.1 bpp WOW and S-UNIWARD variants to the P2 pool at the same 360 pairs, so fewer training pairs carry the stronger 0.4 bpp signal. On unseen HILL 0.4, SRM+Ensemble scores 0.653 under P2 and 0.632 under P3 (ΔAUC = −0.022, 95% CI [−0.050, +0.005]); for Residual-LR the difference is −0.005 [−0.020, +0.007]. Both intervals include zero, so the data show neither an improvement nor equivalence of the two protocols. For SRM+Ensemble the point estimates also fall on matched 0.4 bpp conditions (WOW 0.713 → 0.686, S-UNIWARD 0.717 → 0.677), and AUC at 0.1 bpp does not rise although 0.1 bpp is now trained on. One untested explanation is that, at fixed size, weak 0.1 bpp examples dilute the 0.4 bpp signal.

Answer to RQ2. No reliable improvement was observed under the evaluated mixed-training protocol. This holds for WOW and S-UNIWARD as training algorithms, two payloads, 360 training pairs and this dataset; larger or differently weighted mixed training sets remain untested.

4.4 RQ3: Payload Shift

Fig. 2 shows ΔAUC between 0.1 and 0.4 bpp for all 18 comparisons, and Table 6 summarizes them per detector and protocol; Supplementary Table S2 lists every interval. All 18 differences are negative and all 18 intervals lie below zero. The

SRM+Ensemble losses are about three times those of Residual-LR, and one Residual-LR interval ends at −0.001 (P3 WOW). As the intervals are unadjusted and share covers and models, the common sign is not 18 independent confirmations.

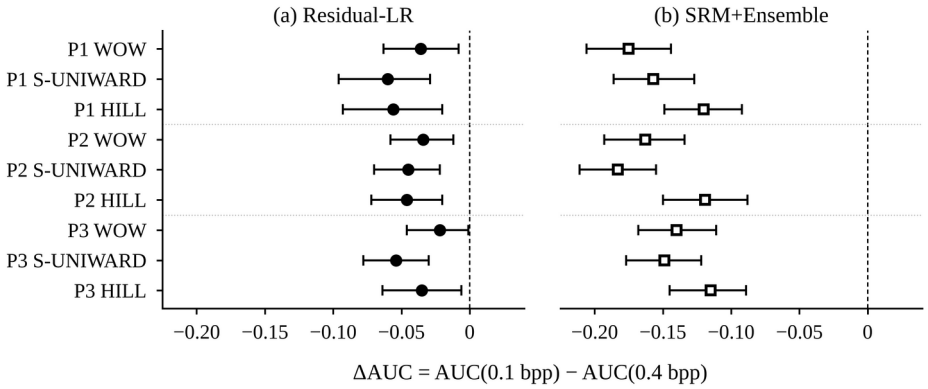


Fig. 2. Payload shift: ΔAUC = AUC(0.1 bpp) − AUC(0.4 bpp) with 95% paired bootstrap intervals. Under P1 and P2 the 0.1 bpp payload is absent from training; under P3 it is included.

Table 6. RQ3 summary over the three algorithms, ΔAUC = AUC(0.1 bpp) − AUC(0.4 bpp), computed from unrounded AUCs.

Detector	P	AUC at 0.1 bpp	ΔAUC range	Mean ΔAUC	Highest CI upper bound	CIs < 0
Residual-LR	P1	0.505-0.514	−0.060 to −0.036	−0.051	−0.008	3/3
	P2	0.500-0.505	−0.046 to −0.034	−0.042	−0.012	3/3
	P3	0.501-0.511	−0.054 to −0.022	−0.037	−0.001	3/3
SRM+Ensemble	P1	0.525-0.539	−0.175 to −0.120	−0.151	−0.092	3/3
	P2	0.534-0.550	−0.183 to −0.119	−0.155	−0.088	3/3
	P3	0.517-0.546	−0.149 to −0.115	−0.135	−0.089	3/3

Answer to RQ3. Lowering the payload from 0.4 to 0.1 bpp lowered the AUC point estimate in all 18 combinations. The loss is clear for SRM+Ensemble, whose AUC of 0.517-0.550 at 0.1 bpp leaves little usable signal, whereas the weakest Residual-LR intervals end close to zero. The loss persists under P3, whose pool contains 0.1 bpp examples, so at 360 pairs training on the lower payload did not offset the weaker signal. The finding is limited to the three algorithms, two payloads and image size studied.

4.5 Security Operating Point at Low FPR

Table 7 reports the operating point of Eq. (1). The threshold allows at most one false positive among the 120 validation covers, yet on test covers every detector-protocol cell exceeds the 1% target, with observed FPRs of 0.022-0.053. Residual-LR under P1 has 15 false positives in 360 test cover evaluations; its three runs are identical (P1 has no variant sampling and the lbfgs fit is deterministic), so that is 5/120 per run, whose exact

95% Clopper-Pearson interval, [1.4%, 9.5%], excludes 1%. A likely reason is that τ_{low} rests on the second-highest of only 120 validation cover scores, an extreme order statistic that varies strongly between samples. An exchangeable test cover exceeds this threshold with probability $2/121 \approx 1.7\%$.

Table 7. Operating point at τ_{low} (validation FPR ≤ 1%). False positives over seeds 42-44 (3 × 120 test covers) are implied by the reported mean test FPR (Suppl. Table S3); recall is the mean and range over the six test conditions.

Detector	P	Test false positives	Observed test FPR	Mean recall	Recall range
Residual-LR	P1	15/360	0.042	0.049	0.042-0.058
	P2	15/360	0.042	0.070	0.053-0.097
	P3	19/360	0.053	0.068	0.056-0.092
SRM+Ensemble	P1	18/360	0.050	0.097	0.053-0.189
	P2	11/360	0.031	0.069	0.025-0.156
	P3	8/360	0.022	0.075	0.019-0.147
Pooled: Residual-LR 49/1,080 (FPR 0.045, mean recall 0.062); SRM+Ensemble 37/1,080 (FPR 0.034, mean recall 0.080)					

Even the best cell, SRM+Ensemble on matched WOW 0.4 under P1, reached a recall of only 0.189 at a test FPR of 0.050. Each detector has 54 evaluation rows (3 protocols × 3 seeds × 6 conditions) and 6,480 stego-score evaluations, but its nine models all score the same 720 test stego images of 120 source covers. These scores are repeated measurements, not independent samples. The false positives in Table 7 are also repeated evaluations of the same 120 test covers. The detectors thus did not demonstrate reliable operation at FPR ≤ 1%, and the results do not support deployment in an operational screening system.

4.6 Discussion and Security Implications

Observed evidence. SRM+Ensemble lost AUC when moving from matched WOW to unseen HILL. This agrees in direction with the largest loss for held-out HILL reported for SRNet [10], although that study trained on three algorithms and about 28,000 images rather than one algorithm and 720 images. Mixing payloads at a fixed training size gave no reliable improvement on HILL; Butora and Fridrich likewise found in preliminary tests that a random mixture of payloads trained no better than a single mid payload [10]. SRM+Ensemble was far weaker at 0.1 bpp and the near-chance Residual-LR slightly weaker, in line with the payload dependence reported for SRM and SRNet [5, 12]. The observed test FPR exceeded the validation target in every cell; [17] likewise cautions that low-FPR estimates from 2,000 test images are noisy, and here the threshold rests on only 120 validation covers.

Possible explanations. The higher intrinsic security of HILL [4], the small training set and the resizing to 256 × 256 may contribute to these AUC losses; none of them was tested here.

For a defender, ranking quality and operational usefulness diverge: an AUC of 0.71 still meant missing over 80% of stego images at a threshold meant to keep false alarms

near 1%. For an adversary, lowering the payload to 0.1 bpp, or using HILL against WOW-trained SRM+Ensemble, was enough to weaken detection. The main limits are the dataset size (600 covers, one split, so the intervals are conditional on that partition), the resizing to 256 × 256, detector strength (two feature-based detectors, one near chance, and no matched HILL reference) and generalization (one cover source, and a 1% threshold calibrated on 120 covers).

5 Conclusion

With the source-cover split fixed before embedding, the three research questions received different answers. For RQ1, SRM+Ensemble trained on WOW at 0.4 bpp detected unseen HILL less well than matched WOW (ΔAUC = −0.070, 95% CI [−0.093, −0.046]), although this loss cannot be separated from HILL being harder to detect. For RQ2, adding 0.1 bpp variants to the training pool gave no reliable improvement on unseen HILL. For RQ3, all 18 comparisons showed lower AUC at 0.1 than at 0.4 bpp, and SRM+Ensemble came close to chance at the lower payload.

These conclusions are limited to 600 BOSSbase covers resized to 256 × 256 and to two feature-based detectors, one of them close to chance. The 1% operating point was calibrated on 120 validation covers and was not met on test covers, and no cross-source data were evaluated.

Future work will use the full BOSSbase at its original resolution, add stronger detectors such as SRNet under the same split, include a matched HILL reference and test cover-source shift. StegaGuard is a controlled evaluation study rather than a new detector, and the results do not indicate readiness for deployment.

Acknowledgement. This research is funded by the University of Economics Ho Chi Minh City, Vietnam (UEH).

References

1. Filler, T., Judas, J., Fridrich, J.: Minimizing additive distortion in steganography using syndrome-trellis codes. IEEE Trans. Inf. Forensics Secur. 6(3), 920–935 (2011). https://doi.org/10.1109/TIFS.2011.2134094
2. Holub, V., Fridrich, J.: Designing steganographic distortion using directional filters. In: IEEE International Workshop on Information Forensics and Security (WIFS 2012), pp. 234–239. IEEE (2012). https://doi.org/10.1109/WIFS.2012.6412655
3. Holub, V., Fridrich, J., Denemark, T.: Universal distortion function for steganography in an arbitrary domain. EURASIP J. Inf. Secur. 2014, 1 (2014). https://doi.org/10.1186/1687-417X-2014-1
4. Li, B., Wang, M., Huang, J., Li, X.: A new cost function for spatial image steganography. In: IEEE International Conference on Image Processing (ICIP 2014), pp. 4206–4210. IEEE (2014). https://doi.org/10.1109/ICIP.2014.7025854
5. Boroumand, M., Chen, M., Fridrich, J.: Deep residual network for steganalysis of digital images. IEEE Trans. Inf. Forensics Secur. 14(5), 1181–1193 (2019). https://doi.org/10.1109/TIFS.2018.2871749

6. Kodovský, J., Fridrich, J., Holub, V.: Ensemble classifiers for steganalysis of digital media. IEEE Trans. Inf. Forensics Secur. 7(2), 432–444 (2012). https://doi.org/10.1109/TIFS.2011.2175919
7. Yedroudj, M., Comby, F., Chaumont, M.: Yedroudj-Net: An efficient CNN for spatial steganalysis. In: IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP 2018), pp. 2092–2096. IEEE (2018). https://doi.org/10.1109/ICASSP.2018.8461438
8. Mallet, A., Beneš, M., Cogranne, R.: Cover-source mismatch in steganalysis: systematic review. EURASIP J. Inf. Secur. 2024, 26 (2024). https://doi.org/10.1186/s13635-024-00171-6
9. Kodovský, J., Sedighi, V., Fridrich, J.: Study of cover source mismatch in steganalysis and ways to mitigate its impact. In: Media Watermarking, Security, and Forensics 2014. Proc. SPIE, vol. 9028, p. 90280J. SPIE (2014). https://doi.org/10.1117/12.2039693
10. Butora, J., Fridrich, J.: Detection of diversified stego sources with CNNs. In: Electronic Imaging, Media Watermarking, Security, and Forensics 2019, pp. 534-1–534-11. IS&T (2019). https://doi.org/10.2352/ISSN.2470-1173.2019.5.MWSF-534
11. Wang, F., Fu, Z., Zhang, X., Li, Z., He, Z., Wang, M.: Pair-wise confidence difference-based pseudo-label selection for universal mismatched steganalysis. In: 33rd ACM International Conference on Multimedia (MM 2025), pp. 8673–8681. ACM (2025). https://doi.org/10.1145/3746027.3755614
12. Fridrich, J., Kodovský, J.: Rich models for steganalysis of digital images. IEEE Trans. Inf. Forensics Secur. 7(3), 868–882 (2012). https://doi.org/10.1109/TIFS.2012.2190402
13. Pevný, T., Bas, P., Fridrich, J.: Steganalysis by subtractive pixel adjacency matrix. In: 11th ACM Workshop on Multimedia and Security (MM&Sec 2009), pp. 75–84. ACM (2009). https://doi.org/10.1145/1597817.1597831
14. Yu, L., Li, Y., Weng, S., Tian, H., Liu, J.: Adaptive multi-teacher softened relational knowledge distillation framework for payload mismatch in image steganalysis. J. Vis. Commun. Image Represent. 95, 103900 (2023). https://doi.org/10.1016/j.jvcir.2023.103900
15. Cogranne, R., Giboulot, Q., Bas, P.: The ALASKA steganalysis challenge: A first step towards steganalysis “into the wild”. In: ACM Workshop on Information Hiding and Multimedia Security (IH&MMSec 2019), pp. 125–137. ACM (2019). https://doi.org/10.1145/3335203.3335726
16. Cogranne, R., Fridrich, J.: Modeling and extending the ensemble classifier for steganalysis of digital images using hypothesis testing theory. IEEE Trans. Inf. Forensics Secur. 10(12), 2627–2642 (2015). https://doi.org/10.1109/TIFS.2015.2470220
17. Butora, J., Bas, P.: MODE: Loss function for deep steganalysis at low false positive rate. In: 33rd European Signal Processing Conference (EUSIPCO 2025), Palermo, Italy (2025). https://hal.science/hal-05101274
18. Bas, P., Filler, T., Pevný, T.: “Break our steganographic system”: The ins and outs of organizing BOSS. In: Information Hiding, IH 2011. LNCS, vol. 6958, pp. 59–70. Springer, Heidelberg (2011). https://doi.org/10.1007/978-3-642-24178-9_5
19. Pedregosa, F., et al.: Scikit-learn: Machine learning in Python. J. Mach. Learn. Res. 12, 2825–2830 (2011)
20. Beneš, M., Lorch, B.: sealwatch: Implementation of modern image steganalysis algorithms. Python package, University of Innsbruck (2024). https://github.com/uibk-uncover/sealwatch